Semiempirical Orbital-Optimized DFT in MOUSE: Spin-Pure Excited-State Surfaces with Analytic Gradients and Hessians

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Abstract

We develop a semiempirical orbital-optimized excited-state model in MOUSE, inspired by Levi and co-workers' direct orbital optimization and electronic saddle searches. A spin-polarized GFN2 energy uses DFT-derived orbital-energy references, fixed 50/50 F/Q weights, exchange-response scale 0.9, and no fitted excitation-energy offset. Optimized spin-pure singlets are coupled to double excitations with a prepared statically screened interaction. A common reflection-even augmentation resolves all seven $\pi\pi^*$ development targets, with 0.180 eV mean absolute error and 0.349 eV maximum error, while retaining thirteen $n\pi^*$ controls within 0.5 eV. The full twenty-state MAE is 0.225 eV, compared with 0.925 eV for the matched native-orbital TDA control. All twenty labels informed this development assessment; the result is not blind validation. Complete analytic gradients and Hessians include orbital stationarity, spin recoupling, CSF metric, moving-ground-projector and state-eigenvector response. A portable MOUSE/LINK implementation follows the coherent spin-pure state without a TDA solve or DFT-backend import after one-time reference preparation. Exact contraction and configuration-block acceleration preserve the differentiated energy. The resulting anchored surface is a candidate for further independent validation of structures, frequencies and chemical transferability.

1 Introduction

The cost of an excited-state calculation is set by the scientific task, not only by the cost of a single vertical energy. Conformer and solvent ensembles, excited-state geometry searches, and local vibrational analyses require repeated evaluations on related molecular structures. A compact, parameterized electronic representation can make these repeated tasks accessible while retaining explicit orbitals, charge redistribution, and a differentiable energy. The purpose of a semiempirical model is therefore to allocate a finite computational budget across the structures and derivatives needed for a molecular prediction. It is not to replace high-level reference calculations indiscriminately. Its usefulness requires a controlled balance of electronic error, nuclear sampling, and computational cost.

The simplified excited-state methods of Grimme and co-workers provide a major reference point. The original sTDA approximates the response matrix and screens the excitation space while retaining a DFT ground-state reference; sTD-DFT extends the simplified construction beyond the Tamm–Dancoff approximation.[1, 2] These are approximations to an excited-state response problem and need not imply a semiempirical ground-state Hamiltonian. In sTDA-xTB, a dedicated tight-binding Hamiltonian and an augmented basis replace the expensive ground-state reference, enabling a predominantly semiempirical spectroscopy workflow.[3] That reference is not interchangeable with an ordinary GFN2 ground-state calculation: the description of the virtual space and the response parameterization are part of the method.

This family has continued to develop. The 2026 std2 description includes sTDA, sTD-DFT, their exact-integral simplified variants XsTDA and XsTD-DFT, and spin-flip formulations, with applications to a broad range of optical response properties.[4] These developments make simplified response methods strong alternatives for multi-root spectroscopy and realistic molecular environments. The present work addresses a complementary objective: an explicitly optimized semiempirical excited-state energy and its analytic nuclear curvature. Its matched GFN2 TDA controls isolate this choice within the MOUSE energy models. Earlier model-generation comparisons with published sTDA-xTB are retained in the Supporting Information; XsTD-DFT is not benchmarked here.

Other low-cost approaches already address excited-state structure and dynamics. TD-DFTB and its long-range-corrected and hybrid extensions combine a compact DFT-derived Hamiltonian with an excited-state response treatment. Recent hybrid-functional DFTB parameterizations target organic photovoltaic molecules, illustrating that frontier levels and charge transfer remain active areas of model development.[5, 6] TDDFT+TB instead keeps a DFT ground state and simplifies the coupling matrix; analytic excited-state gradients were reported in 2023.[7] The approximate auxiliary-function TDDFT-aas approach added an alternative efficient response construction with analytic gradients in 2024.[8] These DFT-reference approaches are relevant cost-reduction strategies, although they do not make the entire calculation semiempirical.

A particularly close precedent is ΔDFTB. Kowalczyk, Le, and Irle implemented state-specific excited-state energies and gradients within DFTB, with benchmarks of geometries, Stokes shifts, and vibrational frequencies.[9] The method remains part of the modern DFTB+ framework.[5] A direct semiempirical excited-state energy is therefore an established idea, not a first introduced here. Orthogonalization-corrected OM2/OM3 and ODM2/ODM3 Hamiltonians combined with multireference configuration interaction (MRCI) provide a different established route to excited-state energies and molecular structure, with a wavefunction expansion rather than a single targeted orbital-optimized CSF.[10, 11] Their treatment of several configurations is especially relevant when evaluating the eventual scope of a ROKS-based model.

The methodological motivation comes from the orbital-optimization studies of Levi and co-workers. Levi, Ivanov, and Jónsson formulated variational excited-state calculations as direct optimization in orbital space, including convergence to electronic saddle points rather than relying exclusively on self-consistent-field iterations.[12, 13] Their work separates the definition of a stationary electronic state from the algorithm used to locate it. Ivanov and co-workers extended this approach to grid and plane-wave representations and discussed excited-state atomic forces.[14] Generalized mode following subsequently provided a way to target a specified saddle order by reversing selected electronic curvature directions.[15] The more recent freeze-and-release strategy uses an initial relaxation with selected excitation orbitals fixed, followed by their release in a full saddle search.[16] Together, these contributions motivate an excited-state solver built around an energy, its orbital gradient, and its curvature.

This perspective is particularly attractive for a semiempirical platform. A ground-state engine already supplies much of the required infrastructure: the energy, Fock matrix, density response, overlap derivatives, and nuclear derivatives. An orbital-optimized excited-state method can reuse these quantities, provided the state-specific orbital equations and their response are added consistently. Reusing the engine therefore means reusing energy and derivative primitives; it does not make an unmodified ground-state Hessian an excited-state Hessian.

MOUSE, the Molecular Orbital Utility for Semiempirical Explorations, supplies a modular environment for such a development. Here its GFN2-based energy is augmented by frozen operators obtained from projected ground-state DFT information. GFN2-xTB provides the semiempirical foundation,[17] while restricted open-shell Kohn–Sham (ROKS) ideas provide a spin-adapted open-shell singlet construction.[18] The resulting method is called *semiempirical orbital-optimized DFT* (OO-DFT) in the sense that a density-functional-response-informed semiempirical energy is made stationary with respect to the excited-state orbitals. It is not a

full-basis Kohn–Sham calculation at each molecular geometry.

Against this background, our contribution is the combination of a GFN2-based, DFT-response-informed energy, a spin-pure ROKS CSF, direct orbital stationarity with state following, and an analytic constrained response through the nuclear Hessian. The contribution concerns this explicit formulation and implementation, rather than priority for semiempirical excited-state gradients or orbital optimization in general. The final examples address vertical accuracy, numerical correctness of the derivatives, and portable state following. Earlier molecular-minimum examples belong to separately identified model generations. TDA is retained as a matched control, using semiempirical rather than DFT orbitals. No general replacement of TDA for all excited-state tasks is implied.

Figure 1 summarizes the flow from frozen reference information and the native semiempirical engine to the stationary excited-state surface and its derivatives.

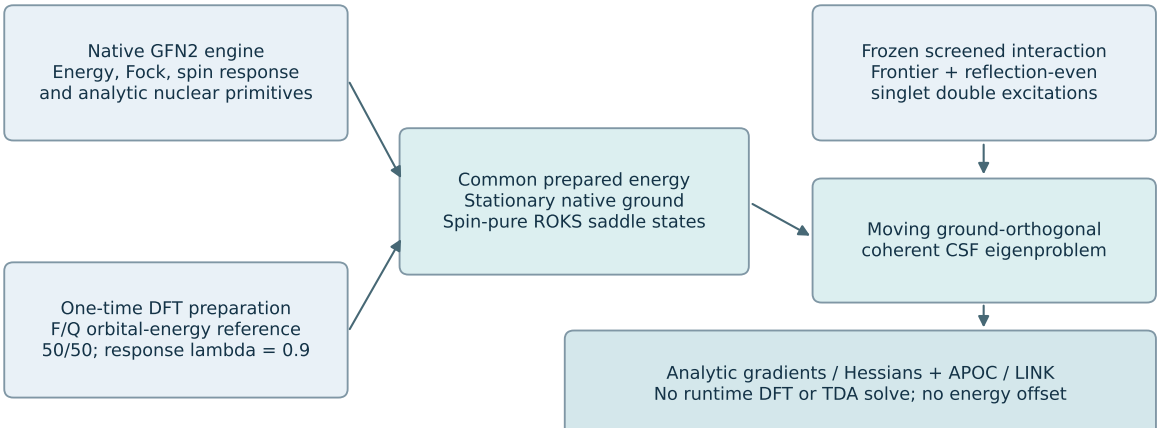


Figure 1: Architecture of the prepared MOUSE OO-CID surface. F/Q information defines one common 50/50 reference before orbital optimization. Native GFN2 energy, spin and derivative primitives support stationary ROKS states. Frozen screened couplings and symmetry-selected singlet configurations define the moving-ground-orthogonal state interaction. Full implicit and eigenvector response gives analytic nuclear derivatives; APOC/LINK retain the coherent spin-pure state.

2 Theory and implementation

2.1 Common reference and spin-polarized native energy

The native GFN2 spatial orbitals C_{n} define the retained valence chart. At the frozen anchor, a core-orthogonal isometry embeds this chart in the DFT AO space. The diagonal Rayleigh quotients of the embedded DFT Fock matrix supply levels F_i. Qualified occupied/virtual assignments transfer DFT orbital spacings to Q_i after aligning occupied centroids; unqualified levels retain F_i. The correspondence thresholds are 0.7 in absolute overlap and 0.05 in best-versus-alternative margin. If the molecular Q transfer loses its occupied–virtual gap, Q reverts to F. The common one-body reference uses $\epsilon_i = (F_i + Q_i)/2$ before orbital optimization. DFT spatial orbitals are not substituted for the native orbitals. This is not an arithmetic average of two independently optimized excited energies.

In a complete symmetric AO-pair moment representation, the prepared energy is

$$E[P^\alpha, P^\beta; R] = E_{\mathrm{native}} + v^{\mathsf{T}} q + \tfrac{1}{2}(q - q_0)^{\mathsf{T}} K_c (q - q_0) + \tfrac{1}{2} m^{\mathsf{T}} K_s m, \tag{1}$$

$$q_a = \mathrm{Tr}[B_a (P^\alpha + P^\beta)], \qquad m_a = \mathrm{Tr}[B_a (P^\alpha - P^\beta)]. \tag{2}$$

E_{native} retains the native electronic, charge-dependent, anisotropic, repulsive, dispersion and spin-polarization terms; K_c and K_s are the prepared additional response corrections. The charge correction subtracts the corresponding native response rather than adding a duplicate Coulomb kernel. The linear term places the reference Fock spectrum in the native chart. The quadratic tensors are prepared at the stationary embedded DFT density and the linear reference is centered on the native ground. Spin polarization is part of the energy and its derivatives, not an additive excitation correction.

The final reference is CAM-B3LYP,[19] with range parameter $0.33\,\mathrm{bohr}^{-1}$. The inherited uracil scale $\lambda = 0.9$ multiplies its exact-exchange response once, giving short- and long-range response coefficients 0.171 and 0.585. The semilocal response is unchanged and orbital gaps are not multiplied by λ. Neither operator construction nor state assignment uses a reference excitation energy.

2.2 Spin purity and electronic saddle stationarity

For a promotion $h \to p$, the two exchanged-spin determinants sharing a single spatial orbital set define a singlet CSF with a consistent fermion phase convention. Its energy is the restricted-open-shell multiplet sum

$$E_S(C, R) = 2E_M(C, R) - E_T(C, R). \tag{3}$$

The singlet CSF is spin pure, whereas the density-functional multiplet-sum energy remains approximate. This does not identify a broken-symmetry $M_s = 0$ determinant as an exact singlet. Four-open-shell occupations require two independent singlet recouplings, both retained below.

The ground is made stationary in the native closed-shell functional. Target ROKS orbitals satisfy the ground-CSF orthogonality constraint and their bordered stationarity equations. Metric-preserving orbital rotations, Newton residual searches, and state-overlap guards locate the assigned branch. An excited solution may be an electronic saddle, as emphasized in the direct-optimization work of Levi and co-workers;[13, 15] no orbital-energy minimum is assumed. Electronic saddle order and nuclear vibrational stability are distinct properties.

2.3 Analytic response of the state-interaction surface

Let z collect the stationary ground and target orbital coordinates and constraint multipliers, and write their residual equations as $F(z(R), R) = 0$. Their first and second analytic responses obey

$$F_z z_a = -F_a, \tag{4}$$

$$F_z z_{ab} = -F_{ab} - F_{az} z_b - F_{bz} z_a - F_{zz}[z_a, z_b]. \tag{5}$$

The bordered linear systems are solved on the regular assigned branch. Both responses are required: the subsequent state-interaction eigenvalue is not stationary with respect to the individually optimized ROKS orbitals. A variational single-determinant Hessian formula alone would omit terms.

For the generalized eigenproblem in the moving ground-orthogonal space, $Hc = \mathcal{E}Sc$ with $c^{\mathsf{T}}Sc = 1$, total matrix derivatives give

$$\mathcal{E}_a = c^{\mathsf{T}}(H_a - \mathcal{E}S_a)c, \tag{6}$$

$$\mathcal{E}_{ab} = c^{\mathsf{T}}(H_{ab} - \mathcal{E}S_{ab})c - \mathcal{E}_b c^{\mathsf{T}} S_a c - \mathcal{E}_a c^{\mathsf{T}} S_b c + 2\sum_{k\neq n} \frac{[c_k^{\mathsf{T}}(H_a - \mathcal{E}S_a)c][c_k^{\mathsf{T}}(H_b - \mathcal{E}S_b)c]}{\mathcal{E} - \mathcal{E}_k}. \tag{7}$$

Here the derivatives include the orbital/multiplier responses, metric, transported operators, and moving ground projector. The relevant metric and eigenvalue separations must be resolved. The absolute excited-state energy defines molecular forces and curvature; vertical excitation energies

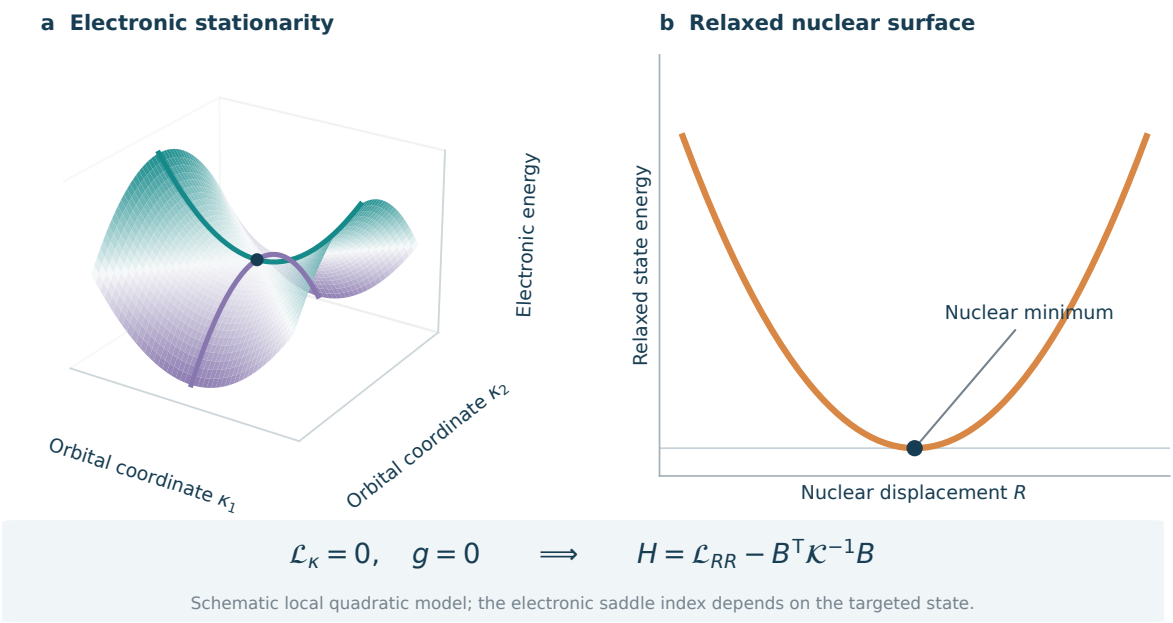


Figure 2: Schematic distinction between electronic and nuclear stationarity. The local quadratic illustration has one unstable orbital direction and a stable relaxed nuclear branch; it is not a molecular calculation. The number of unstable orbital directions need not be one. The displayed quadratic Schur-complement illustration does not replace the full CI response below.

subtract the stationary native ground. Finite differences enter only the independent validation of these analytic derivatives.

3 Computational protocol

The final assessment contains twenty valence transitions in eighteen molecules: seven $\pi\pi^*$ and thirteen $n\pi^*$ targets. Reference energies are the archived QUEST theoretical best estimates, with individual method labels retained in the input data.[20, 21] Molecular geometries and target identities are unchanged from the frozen inputs. All twenty numerical labels were exposed during development of the final augmentation; none is presented as an independent blind test.

The prepared CAM-B3LYP references retain the archived aug-cc-pVDZ AO basis, numerical grid level 3, frozen core embedding, and native valence space. Archived DFT transition information supplies assigned-state seeds and character checks in this benchmark. It does not supply excitation-energy shifts and is not read during prepared runtime. Preparation, including this reference information, remains distinct from evaluation of the MOUSE surface. Completed references, stationary states and operators are reused.

Earlier reference calculations were performed on `oracle.sns.it`; the final configuration-space development and derivative validations were run on the separately verified MAC-STUDIO host. No production electronic calculation was run on the laptop. Runtime property checks use two BLAS threads per process. The parallel nuclear-column validation limits concurrent processes explicitly and preserves every completed column checkpoint.

Matched TDA controls use the same stationary native GFN2 spatial orbitals, F/Q one-body preparation, density-response kernel and $\lambda = 0.9$:

$$A_{ia,jb} = (\epsilon_a - \epsilon_i)\delta_{ij}\delta_{ab} + K_{ia,jb}. \tag{8}$$

The complete nonlocal singlet A operator is used, including the exchange contribution missing from a real symmetrized static response. These controls are distinct from published sTDA-xTB

and from a separately optimized TDA parameterization. The screened interaction introduced below is an additional ingredient of OO-CID, not a hidden modification of the reported TDA values.

4 Screened, symmetry-augmented optimized-target CID

The final development candidate addresses the two remaining $\pi\pi^*$ outliers by changing the configuration space and its interaction, while retaining the native GFN2 orbitals, the same prepared F/Q one-body reference, 50/50 weights, $\lambda = 0.9$, and zero excitation-energy offset. It combines optimized ground-orthogonal ROKS target singlets, their paired doubles, and the original two-occupied/two-virtual frontier doubles in both reflection parities. The added configurations are all spin-singlet doubles from the five highest occupied and five lowest virtual canonical native orbitals that are even under reflection in the molecular plane. Both independent singlet recouplings of every four-open-shell occupation are included. Canonical paired doubles with retained-CSF overlap at least 0.7 are omitted by the unchanged duplicate-removal rule. An exact Gaussian-overlap representation certifies the nuclear reflection and canonical MO parities at preparation. The resulting configuration list is frozen during subsequent nuclear displacement; it is not reclassified when an out-of-plane coordinate changes. If reflection or MO parity cannot be certified, no additional symmetry-sector configuration is selected.

This is one common CSF space and one effective matrix for all requested states of a molecule. No $n\pi^*/\pi\pi^*$ label, reference excitation energy, molecular exception, or fitted damping factor enters the selection. For the planar examples, $\pi\pi^*$ targets are reflection-even, whereas $n\pi^*$ targets are reflection-odd. The symmetry augmentation therefore adds the missing even-sector flexibility without introducing the large odd-sector manifold that overcorrelated several controls in the unrestricted five-pair diagnostic. The choice was made after inspecting the exposed twenty-state development set and was frozen in commit `2d59ab8`. It is a model-development choice, not blind validation or a general prescription for nonplanar molecules, CT, or Rydberg excitations.

Environmental screening is introduced through a static finite-basis partition motivated by constrained RPA.[22] Let V be the bare Coulomb matrix in the orthonormal AO-pair chart, Δ the positive native occupied–virtual gap matrix, and T the symmetrized transition-density columns weighted by their environmental fraction. With occupied and virtual active projectors P_h, P_v, these fractions are $1 - (P_h)_{ii}(P_v)_{aa}$; the projectors span the five-pair frontier and retained optimized hole/particle directions. Our prepared static interaction is

$$W = V - VT(\Delta + T^{\mathsf{T}}VT)^{-1}T^{\mathsf{T}}V. \tag{9}$$

This explicitly specified finite-basis approximation is not a full frequency-dependent cRPA calculation. The same W acts in both reflection sectors. Its one-body term is normal ordered to the unchanged native reference Fock matrix, $h_W = F_{\mathrm{ref}} - J_W[P_g] + K_W[P_g]/2$. Let H^W_{IJ} and S_{IJ} denote its Slater–Condon matrix and CSF metric. The native spin-polarized functional supplies each diagonal energy E_I, and the effective matrix is

$$H_{IJ} = H^W_{IJ} + \tfrac{1}{2}S_{IJ}(d_I + d_J), \qquad d_I = E_I - H^W_{II}/S_{II}. \tag{10}$$

Every two-dimensional four-spin singlet block is restored to the exact SU(2) completion of the same quadratic spin functional. Excited CSFs are projected orthogonal to the moving stationary ground CSF before solving the generalized eigenproblem. Excitation energies subtract its native stationary energy, not an auxiliary correlated-ground eigenvalue. The overlap-weighted residual remains an approximate multistate functional; it is not a proof of a universally balanced correlated ground-state theory.

All seven $\pi\pi^*$ targets now satisfy the unchanged 0.5 eV individual criterion: their MAE is 0.180 eV and maximum absolute error is 0.349 eV. Furan changes from 5.804 to 6.609 eV against

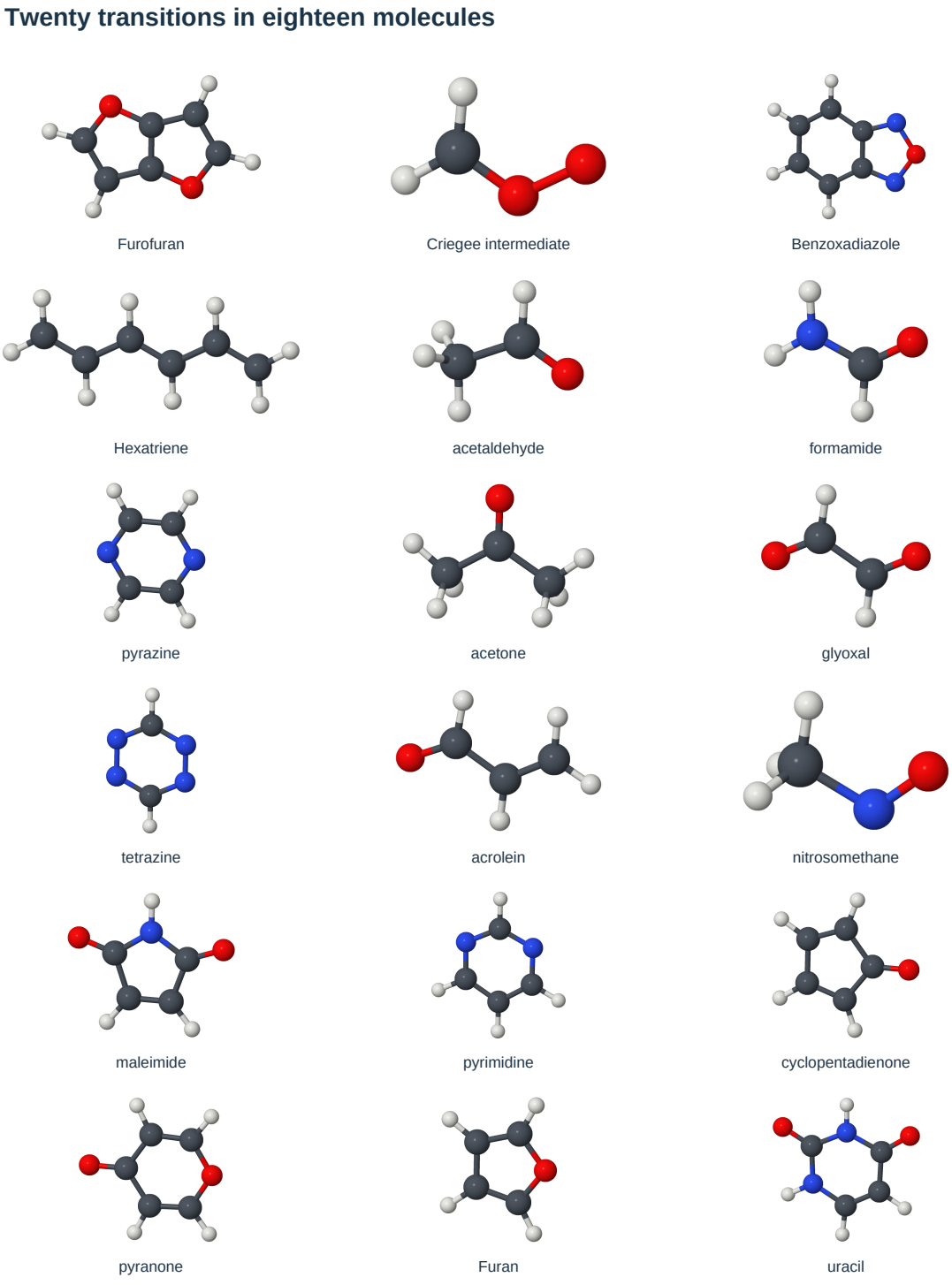


Figure 3: The eighteen molecular anchors used for the final twenty-state assessment, rendered with Jmol from the archived three-dimensional coordinates. Only rigid display rotations are applied; panel scales differ. Furofuran contributes two $\pi\pi^*$ states, and Criegee contributes one state of each valence class.

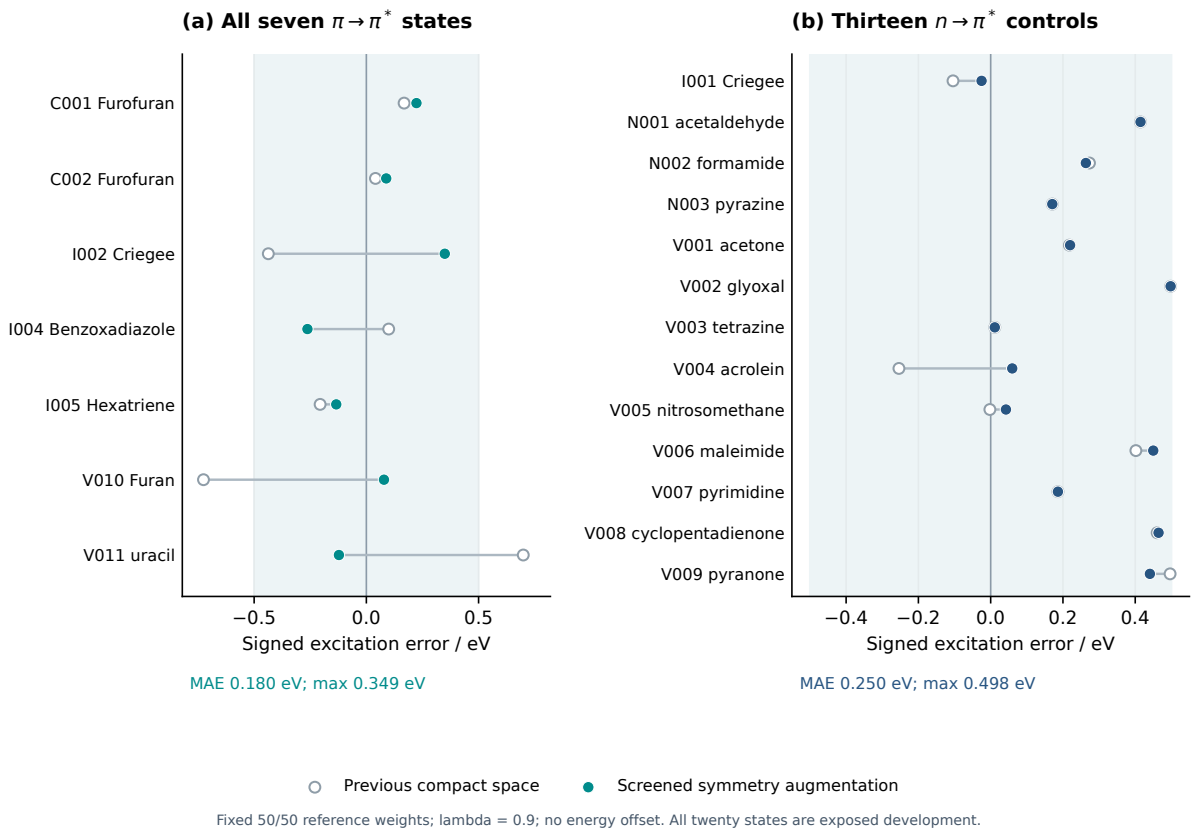


Figure 4: Resolution of the seven $\pi\pi^*$ development targets while retaining the thirteen $n\pi^*$ controls. Open circles show the previous compact-space errors; filled circles show the screened symmetry augmentation. The shaded interval is ±0.5 eV. All twenty states informed the development assessment; this figure is not a blind-validation result.

6.530 eV, and uracil from 6.080 to 5.259 eV against 5.381 eV. The thirteen $n\pi^*$ controls remain within 0.5 eV, with 0.250 eV MAE and 0.498 eV maximum error. Overall, the twenty states give 0.225 eV MAE. These are exposed-data development results: the earlier post-freeze designation does not make them prospective tests of this newly selected model. The matched native-orbital TDA control retains its 0.925 eV MAE, using the same F/Q preparation and native response kernel; the new screened interaction is an additional ingredient of OO-CID. This comparison does not establish superiority to every calibrated TDA, sTDA-xTB, or TD-DFTB parameterization. All individual values are in the Supporting Information.

The public `PreparedNOSIModel` accepts a checksummed extended model, and LINK retains its `nosi_50_50` selector and accepted-restart contract. Runtime evaluation reuses only ground-engine Fock, energy, spin-polarization, and derivative primitives; it performs no TDA solve, DFT-backend import, or new SCC calculation after preparation. APOC follows the coherent determinant sum and verifies its norm and $\langle S^2 \rangle = \|S_+\Psi\|^2 = 0$. An individual broken-symmetry $M_s = 0$ determinant is never identified as a spin-pure singlet. The analytic response differentiates the native ground and target KKT systems, all diagonal functionals and spin recouplings, the CSF metric, generalized eigenvectors, and moving ground projector. The prepared h_W, W, auxiliary orbitals, and configuration list are fixed in the co-rotating orthonormal AO frame. The resulting derivatives are exact for this anchored surface; rebuilding the DFT reference or screening at each geometry defines a different model.

Canonical auxiliary blocks are contracted exactly with frozen-core occupation bitstrings, while optimized-state cross terms retain the singular-safe nonorthogonal Schur/cofactor algebra. Further acceleration caches NumPy's greedy contraction programs, omits only exactly zero

derivative terms, and reuses the invariant auxiliary Gram block. Neither an integral nor a derivative term is approximated. Independent canonical-block comparisons agree below 10^{-12} Hartree; a molecular mixed-response column agrees with the original implementation within 4.2×10^{-15} Hartree/bohr^2. The extended public surface and response are committed as `81897cf`; the exact contraction and metric acceleration is `e572d50`. Complete analytic Hessians are qualified for Criegee (two states, 15 Cartesian coordinates), furan (27 coordinates), and uracil (36 coordinates). Independently reconverged energies at 10^{-3} and 5×10^{-4} bohr validate both an out-of-plane direction and a mixed internal displacement. Richardson-extrapolated maximum errors are 1.42×10^{-10} Hartree/bohr for first derivatives and 3.71×10^{-7} Hartree/bohr^2 for curvatures. The largest raw Hessian asymmetry is 4.64×10^{-14} Hartree/bohr^2, and the maximum translational identity residual is 1.43×10^{-13} Hartree/bohr^2. These are implementation checks of the final anchored surface, not a validation of its equilibrium geometries or physical vibrational frequencies. Fresh-process runtime, spin and restart checks pass for all twenty states in eighteen molecules. Prepared energy-only evaluations take 1.7–9.0 s with two BLAS threads on MAC-STUDIO, excluding model loading and reference preparation. They do not demonstrate a vertical speed advantage over TDA. A direct cold public-API Criegee Hessian reproduces the independently assembled matrices within 2.2×10^{-9} Hartree/bohr^2. Its complete two-state Hessian evaluation takes 418 s with two BLAS threads and about 1.0 GiB peak RSS; this is the current dense reference implementation, not an end-to-end timing advantage over TDA. The separate column-checkpoint route permits bounded remote parallel validation and reuses completed columns. The final LINK $\pi\pi^*$ test has restart error 3.6×10^{-15} Hartree and displaced trial overlap 0.99999996; trial evaluation leaves the accepted checkpoint unchanged.

5 Position, cost, and remaining validation

The rationale for the semiempirical representation is repeated molecular work with an explicit differentiable energy. Ground-engine primitives support orbital relaxation, coherent state tracking, gradients and nuclear curvature in the same prepared model. This complements the efficient multi-root response and optical-property goals of Grimme's simplified methods.[1, 3, 4] It does not establish an advantage for transition intensities or high-throughput multi-root spectroscopy.

TD-DFTB, TDDFT+TB and TDDFT-aas already offer important low-cost excited-state and gradient capabilities.[5, 7, 8] ΔDFTB supplies an especially close state-specific precedent, including molecular structures and vibrational properties.[9] OMx/ODMx–MRCI provides an established multireference semiempirical alternative.[10, 11] The contribution here is the explicit GFN2/DFT-response-informed energy, spin-pure orbital-optimized configurations, coherent state interaction, and full analytic response through the nuclear Hessian. It is not priority for semiempirical excited-state gradients. A documented TD-DFTB vibronic implementation obtains the excited-state Hessian by differences of analytic gradients;[23] this specific comparison is not a claim that analytic second derivatives are absent from every competing implementation.

The cost accounting is

$$T_{\text{workflow}} = T_{\text{reference}} + T_{\text{projection/screening}} + \sum_{k=1}^{N_{\text{geom}}} T_{\text{OO/CI/derivatives}}(R_k). \tag{11}$$

Reference preparation can be amortized, but its cost does not disappear. The dense complete AO-pair correction and frozen interaction can require fourth-power storage in the retained AO dimension. The current single-root TDA vertical controls are faster than the prepared OO-CID evaluations; the measured advantage on this development set is accuracy and an explicit state-specific differentiated energy, not a demonstrated vertical speedup. An all-inclusive final-model size-scaling benchmark remains separate from the derivative-correctness checks. Earlier model-generation comparisons with published sTDA-xTB, molecular minima, frequencies and cost data are retained in the Supporting Information and are not attributed to this final model.

The next scientific assessment should freeze this complete preparation and space-selection rule before opening new labels, include additional valence, CT and Rydberg states, and compare final-model excited minima and frequencies with suitable high-level or experimental references. Sensitivity to anchor geometry and larger nuclear displacements also remains to be established. The present finite planar-state result does not demonstrate universal transferability or replace those tests.

6 Conclusions

MOUSE provides a semiempirical orbital-optimized route with explicit spin polarization, spin-pure singlet CSFs, and analytic nuclear gradients and Hessians. Ground-program Fock, energy and derivative primitives are reused through constrained electronic saddle optimization and implicit response. The final screened OO-CID model retains the original frontier space in both reflection parities and adds its larger reflection-even valence sector. The F/Q weights remain 50/50, the exchange-response scale remains 0.9, and no excitation-energy offset is introduced.

All seven $\pi\pi^*$ development targets now meet the fixed individual accuracy criterion, with 0.180 eV MAE and 0.349 eV maximum error. The thirteen $n\pi^*$ controls also remain within 0.5 eV; all twenty states give 0.225 eV MAE. These results resolve the previous furan and uracil outliers on this finite set. They do not establish universal accuracy: all twenty numerical labels were exposed during model development, and the symmetry augmentation has been assessed here on planar valence states.

The public surface retains coherent APOC state following and accepted-only LINK restarts, and uses the ground-state engine without a runtime TDA or DFT calculation. Its analytic response differentiates the defined anchored model, including the moving ground projector and all state-interaction terms. Earlier production-model minima and frequency comparisons remain separately identified. A broader independent test, final-model excited-state structures and frequencies, and direct comparisons to additional calibrated low-cost methods remain necessary before claiming general predictive performance or a replacement for multi-root TDA spectroscopy.

Data and software availability

MOUSE source is hosted at `https://github.com/yogibubu/MOUSE`. The final symmetry-augmentation choice is frozen in `2d59ab8`; the portable surface and analytic response are implemented in `81897cf`, with exact contraction/metric acceleration in `e572d50`. Final qualification and its complete logs are committed in `358ae21` and `7c5c7f9`. The qualification directory `qualification/pi_pi_transfer_repair_20261101` contains the complete twenty-state predictions, model/exposure protocol, remote runtime and spin checks, analytic-column checkpoints, independent energy validations and executed-source hashes. The manuscript package contains the machine-readable results, scientific figure sources, archived Jmol renders and coordinate provenance. Earlier generations and their qualification archives are identified separately in the Supporting Information and preserved development manuscript. A permanent public dataset identifier has not yet been assigned.

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